

INTRODUCTION TO APPLIED NUMERICAL METHODS

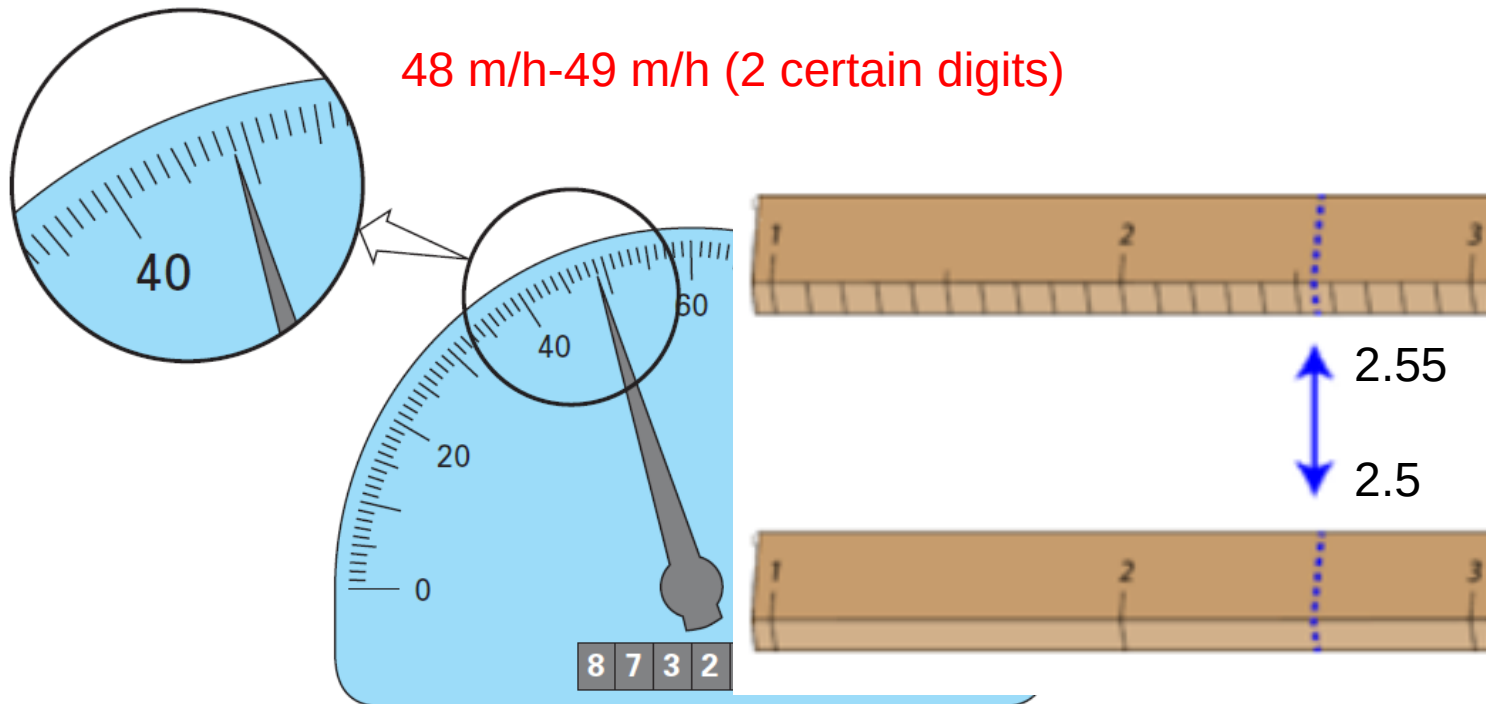
Taylor Series, Truncation Error & Error Propagation

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SIGNIFICANT FIGURES: APPROXIMATE REPRESENTATION OF NUMBERS

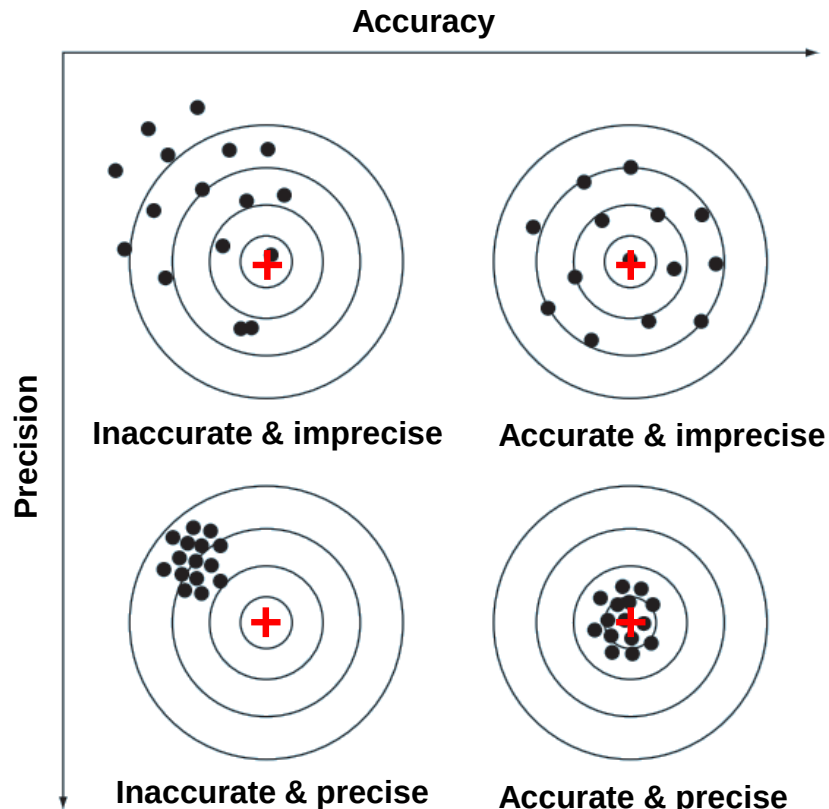
- The significant figures/digits of a number are those that can be used with confidence
- They correspond to the number of certain digits plus one estimated digit
- It is conventional to set the estimated digit at one-half of the smallest scale division on the measurement device



- Zeros are not always significant figures as they may be necessary to just locate a decimal point
- What is the number of the Significant figures for the following sets?
- 0.00112; 0.0112
- 10,000
- 1.0E+4, 1.00E+4

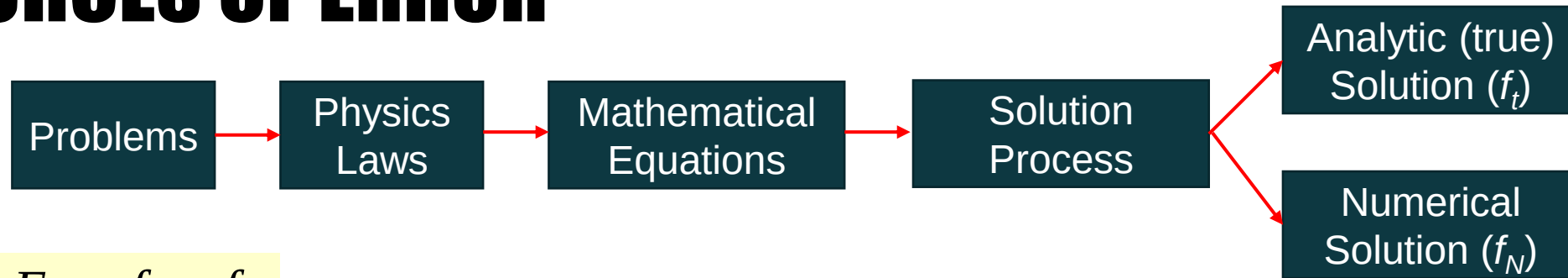
CHARACTERIZATION OF ERROR

- Accuracy: How closely a data point (computed or measured value) agrees with the **true value**
 - Inaccuracy/bias



- Precision: How closely individual data points (computed or measured values) agree with **each other**
 - Imprecision/uncertainty
- Error: including both inaccuracy (bias) and imprecision

SOURCES OF ERROR



$$\text{Error: } E_t = f_t - f_N$$

True value Numerical value

$$\text{Error: } E_t = E_{trunc} + E_{round-off} \quad (\text{True error})$$

- **Round-off error** $E_{round-off}$: computers can represent only quantities with a finite number of digits
- **Truncation error** E_{trunc} : discrepancy introduced by the fact that numerical methods employ approximations to represent exact mathematical operations and quantities

$$\text{True Fractional Relative Error: } \varepsilon_t = \left| \frac{f_t - f_N}{f_t} \right| \times 100\%$$

Relative to True value

AN EXAMPLE OF CALCULATION OF ERRORS

Length	f_t	f_N	E_t	ϵ_t
Rivet	10 cm	9 cm	1 cm	$\epsilon_t = \frac{1}{10}100\% = 10\%$
Bridge	10,000 cm	9,999 cm	1 cm	$\epsilon_t = \frac{1}{10,000}100\% = 0.01\%$

$$\text{True Fractional Relative Error: } \epsilon_t = \left| \frac{f_t - f_N}{f_t} \right| \times 100\%$$

Same absolute error can yield significantly different relative error!

APPROXIMATE ERRORS

- Issue with true error ε_t : we will not know true value/answer f_t *a priori*
- Approximate error
 - Normalize the error using the best available estimate of the true value

$$\text{Approx. Relative Error: } \varepsilon_a = \left| \frac{f_a - f_N}{f_a} \right| \times 100\%$$

Relative to estimate of true value

- Without knowledge/estimate of true value, we evaluate the difference between successive iterations (iterative approach)
 - (we hope) Approximation will become better and better, and ultimately approaches true values

"k" current approximation "k" previous approximation

$$\text{Approx. Relative Error: } \varepsilon_a = \left| \frac{f_a^k - f_a^{k-1}}{f_a^k} \right| \times 100\%$$

number of significant figures

Pre-specified Percent Tolerance

$$\varepsilon_a < \varepsilon_s$$

$$\varepsilon_s = (0.5 \times 10^{2-n})\%$$

Engineering
the result is correct to at least n significant figures

AN EXAMPLE OF APPROXIMATE ERROR

Approximate the exponential function @ $x = 0.5$ using the Maclaurin Series expansion

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!}$$

Such that the absolute value of the approximate error estimate ε_a falls below a pre-specified error criterion ε_s conforming to three significant figures ($n = 3$)

Solution: True value: $e^{0.5} = 1.648721 \dots$

Step 0: determine error criterion corresponding to three significant figures

$$\varepsilon_s = (0.5 \times 10^{2-3})\% = 0.05\%$$

Step 1: A single term $e^x = 1$

$$\varepsilon_t = \frac{1.648721 - 1}{1.648721} \times 100\% = 39.3\%$$

$$\varepsilon_a = \text{N/A}$$

Step 2: Two terms $e^x = 1 + x \Rightarrow e^{0.5} = 1 + 0.5 = 1.5$ for $x = 0.5$

$$\varepsilon_t = \frac{1.648721 - 1.5}{1.648721} 100\% = 9.02\%$$

$$\varepsilon_a = \frac{1.5 - 1}{1.5} 100\% = 33.3\%$$

because $\varepsilon_a > \varepsilon_s$, the process continues ...

AN EXAMPLE OF APPROXIMATE ERROR

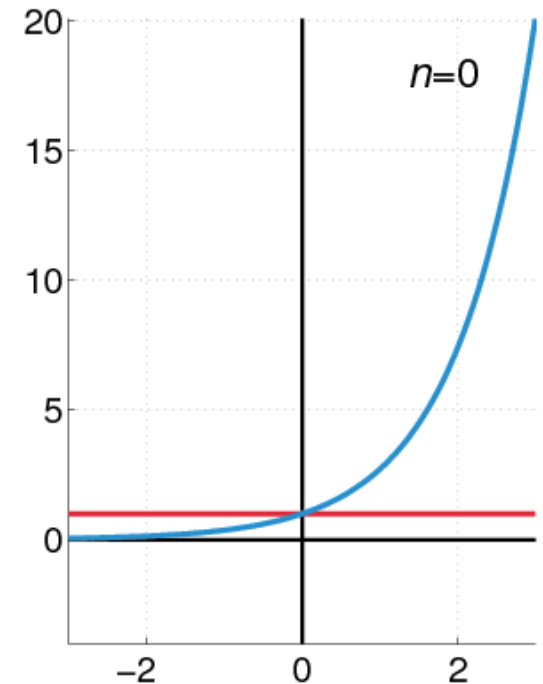
Number of Terms	Numerical Approximation	ε_t (%)	ε_a (%)
1	1	39.3	
2	1.5	9.02	33.3
3	1.625	1.44	7.69
4	1.645833333	0.175	1.27
5	1.648437500	0.0172	0.158
6	1.648697917	0.00142	0.0158

True value: 1.648721 ...

Accurate up to 5 significant figures !!!

$\varepsilon_s = (0.5 \times 10^{2-n})\%$ is conservative for this case

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!}$$



By Oleg Alexandrov - self-made with MATLAB., Public Domain,
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TAYLOR SERIES

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!}$$

$$f(x_{i+1}) = f(x_i) + f'(x_i)(x_{i+1} - x_i) + \frac{f''(x_i)}{2!}(x_{i+1} - x_i)^2 + \frac{f^{(3)}(x_i)}{3!}(x_{i+1} - x_i)^3 + \dots + \frac{f^{(n)}(x_i)}{n!}(x_{i+1} - x_i)^n + R_n$$

Remainder

Why Taylor series?

It provides a means to predict a function value at one query point (x_{i+1}) in terms of the function value $f(x_i)$ and its derivatives $f^{(n)}(x_i)$ at an expansion point (x_i)

Truncation Error?

Approximations by **a finite number of terms** (e.g., $n + 1$ terms) to represent exact function values $f(x_{i+1})$ and it is **a polynomial** (any smooth function can be approximated as a polynomial)

R_n : Remainder - the terms that are truncated due to the approximation

$$R_n = \frac{f^{(n+1)}(x_i)}{(n+1)!}(x_{i+1} - x_i)^{n+1} + \frac{f^{(n+2)}(x_i)}{(n+2)!}(x_{i+1} - x_i)^{n+2} + \dots \quad \text{or} \quad R_n = \frac{f^{(n+1)}(\xi)}{(n+1)!}h^{(n+1)}$$

$$R_n = O\left((x_{i+1} - x_i)^{n+1}\right)$$

The Taylor Series is the “ n^{th} order approximation”

TAYLOR SERIES

For nonlinear function $f(x)$, if it can be expressed as a polynomial, we have:

$$f(x) = c_0 + c_1(x - a) + c_2(x - a)^2 + c_3(x - a)^3 + \dots$$

With a choose value “ a ”, we need to find out values for $c_0, c_1, c_2, \dots$

This is done with the following derivatives:

To get c_0 : If $x = a, f(a) = c_0$

To get c_1 : take the derivative of $f(x)$

$$f'(x) = c_1 + 2c_2(x - a) + 3c_3(x - a)^2 + \dots$$

If $x = a, f'(a) = c_1$

To get c_2 : take the derivative again:

$$f''(x) = 2c_2 + 3 * 2 * c_3(x - a) + \dots$$

If $x = a, f''(a) = 2c_2$, so $c_2 = \frac{f''(a)}{2}$

Continue the derivatives, you will get $c_3, c_4 \dots$

Overall

$$f(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2}(x - a)^2 + \frac{f'''(a)}{3!}(x - a)^3 + \dots$$

Replace x with x_{i+1} and a with x_i , you will have the normal Taylor series.

TAYLOR SERIES: ANOTHER FORM

Step size: $h = x_{i+1} - x_i$

Query
point

Expansion
point

$$f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)}{2!}h^2 + \frac{f^{(3)}(x_i)}{3!}h^3 + \dots + \frac{f^{(n)}(x_i)}{n!}h^n + R_n$$

Function
Values @
expansion
points

Derivative
Values @
expansion
points

Remainder

TAYLOR SERIES: AN EXAMPLE

Use 0th through 4th Taylor Series expansions to approximate

$$f(x) = -0.1x^4 - 0.15x^3 - 0.5x^2 - 0.25x + 1.2$$

Computing function's value at $x_{i+1} = 1$ by expanding the series at $x_i = 0$

Solution:

Query point: $x_{i+1} = 1$; Expansion Point: $x_i = 0$; step size $h = 1$

$$f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)}{2!}h^2 + \frac{f^{(3)}(x_i)}{3!}h^3 + \dots + \frac{f^{(n)}(x_i)}{n!}h^n + R_n$$

$$\text{true value: } f(1) = 0.2$$

$$0^{\text{th}} \text{ Order } (n = 0): \quad f(x_{i+1}) = f(1) \approx f(0) = 1.2 \quad \text{true err: } E_t = 0.2 - 1.2 = -1.0$$

$$1^{\text{st}} \text{ Order } (n = 1): \quad f'(x) = -0.4x^3 - 0.45x^2 - 1.0x - 0.25 \quad \Rightarrow \quad f'(0) = -0.25$$

$$f(x_{i+1}) = f(1) \approx f(0) - 0.25h = 0.95$$

$$E_t = 0.2 - 0.95 = -0.75$$

TAYLOR SERIES: AN EXAMPLE

Use 0th through 4th Taylor Series expansions to approximate

$$f(x) = -0.1x^4 - 0.15x^3 - 0.5x^2 - 0.25x + 1.2$$

Computing function's value at $x_{i+1} = 1$ by expanding the series at $x_i = 0$

Solution:

Query point: $x_{i+1} = 1$; Expansion Point: $x_i = 0$; step size $h = 1$

$$f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)}{2!}h^2 + \frac{f^{(3)}(x_i)}{3!}h^3 + \dots + \frac{f^{(n)}(x_i)}{n!}h^n + R_n$$

$$\text{true value: } f(1) = 0.2$$

$$f'(x) = -0.4x^3 - 0.45x^2 - 1.0x - 0.25$$

$$2^{\text{nd}} \text{ Order } (n = 2): \quad f''(x) = -1.2x^2 - 0.9x - 1.0 \quad \Rightarrow \quad f''(0) = -1.0$$

$$f(x_{i+1}) = f(1) \approx f(0) - 0.25h - \frac{1}{2!}h^2 = 0.45$$

$$E_t = 0.2 - 0.45 = -0.25$$

TAYLOR SERIES: AN EXAMPLE

Use 0th through 4th Taylor Series expansions to approximate

$$f(x) = -0.1x^4 - 0.15x^3 - 0.5x^2 - 0.25x + 1.2$$

Computing function's value at $x_{i+1} = 1$ by expanding the series at $x_i = 0$

Solution:

Query point: $x_{i+1} = 1$; Expansion Point: $x_i = 0$; step size $h = 1$

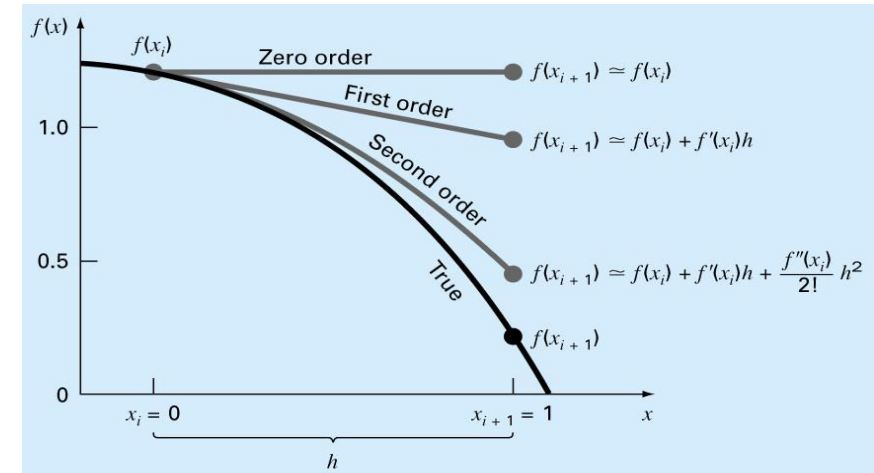
$$f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)}{2!}h^2 + \frac{f^{(3)}(x_i)}{3!}h^3 + \dots + \frac{f^{(n)}(x_i)}{n!}h^n + R_n$$

4th Order ($n = 4$): $f^{(4)}(0) = -2.4$

$$f(x_{i+1}) = f(1) \approx f(0) - 0.25h - \frac{1}{2}h^2 - \frac{0.9}{3!}h^3 - \frac{2.4}{4!}h^4 = 0.2$$

$$E_t = 0$$

$$R_4 = O(h^5) = 0$$



true value: $f(1) = 0.2$

TAYLOR SERIES: AN EXAMPLE

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$
$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

- Technical Findings
 - The n^{th} -order Taylor series expansion will be exact for an n^{th} -order polynomial (derivatives of higher order are all zeros and $R_n = 0$)
 - For general functions (differentiable and continuous), such as exponentials (exp) and sinusoids (sin, cos), a finite number of terms will never produce an exact estimate.
 - In practice, the inclusion of only a few terms could yield an approximation close enough to the true value
 - We can also use the step size h to control the truncation error

Remainder $R_n = O(h^{n+1})$

$R_0 = O(h) \sim h$	Halving step size will halve the error
$R_1 = O(h^2) \sim h^2$	Halving step size will quarter the error

R_n is very useful to evaluate the **comparative error** of numerical methods based on Taylor series expansions